

Mohamednur Zeinu

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Portfolio: <https://mohamednurz.github.io/portfolio/>

ACADEMIC & CO-OP STATUS

Electrical Engineering, BASc Co-op, University of Alberta

Cumulative Grade Point Average

Completed Academic Terms

Completed Co-op Work Terms

Availability Starting January 2027

Class of 2029

3.2/4.0

4 of 8

0 of 4

8 months

TECHNICAL SKILLS

EQUIPMENT

Oscilloscope

Soldering (SMT/THT)

Breadboarding

Hardware Debugging

DESIGN

Altium Designer

KiCad

Vivado

PROGRAMMING

Python

C / C++

Assembly (ARM)

SystemVerilog, VHDL

PRODUCTIVITY

MS Office (Word, Excel & PowerPoint)

WaveForms

STM32CubeMX

PROJECT EXPERIENCE

16-Bit CPU Design

July 2026 – Aug 2026

Personal Project

- Designed a custom 16-bit CPU instruction set architecture (ISA) spanning 16 instructions across 5 encoding formats, defining a complete register-based, load/store processor from scratch.
- Implemented 9 SystemVerilog hardware modules forming a full single-cycle data path, including a 4096-word instruction memory and 4096-word data memory with memory-mapped I/O.
- Programmed two-pass Python assembler, translating custom assembly into 16-bit machine code. Then verified CPU functionality by executing a 12-instruction Fibonacci sequence program, resulting in correct execution with zero errors.
- Collaborated with a project partner on hardware and board-level constraints, jointly reviewing and debugging synthesis, timing, and pin-mapping issues to prepare the design for physical FPGA deployment.
- Synthesized and deployed the complete processor onto a Xilinx Zynq-7000 FPGA, achieving 125 MHz clock frequency while utilizing 0.1% of available FPGA logic resources.

Flight Controller (Project Maverick)

May 2025 – July 2025

Personal Project

- Designed a custom drone flight-controller PCB from scratch in Altium Designer with schematic capture, component selection, and full PCB layout.
- Selected and placed 86 components across 47 unique BOM line items, creating 14 custom PCB footprints (5 built to IPC standard) for parts with no stock library match.
- Integrated an STM32F405RGT6 ARM Cortex-M4 MCU (168 MHz) with a BMI088 IMU, BMP388 barometer, magnetometer, EEPROM, NAND flash, USB, SBUS input, and PWM motor driver.
- Ran a Design Rule Check across 13 constraint categories on the completed 86-component layout, achieving 0 short-circuit violations and 100% net routing completion (0 unrouted nets).
- Built a real-time flight-stabilization control system in Python, fusing IMU accelerometer and gyroscope data with a complementary filter (98% gyroscope / 2% accelerometer) to estimate pitch/roll orientation at a 50 Hz loop rate.

ADDITIONAL WORK EXPERIENCE

Radio Frequency (RF) Communications Team Member

Sep 2025 – Present

AlbertaSat, Edmonton, AB

- Evaluate radio frequency hardware architectures by analyzing system specifications and conducting trade studies using an Antenna Decision Matrix to optimize RF component selection for satellite operations.
- Assist with tracking amateur radio examination data and licensing records for team members to maintain operational compliance.

Automotive Service Advisor

Sep 2025 – Present

Canadian Tire, Edmonton, AB

- Conduct structured vehicle diagnostic interviews and translate technical findings into clear service recommendations, coordinating repair orders between customers and licensed automotive technicians.
- Manage concurrent service desk operations including inbound calls, walk-in consultations, and multi-vehicle status tracking with more than 25 vehicles a day on average resulting in \$300K+ average monthly revenue.
- Monitor repair timelines, proactively communicate vehicle status and updates to customers.

Senior Automotive Service Advisor

Apr 2023 – Sep 2024

Canadian Tire, Fort McMurray, AB

- Facilitated 20+ daily customer service interactions by scheduling appointments, processing warranty claims worth \$30K+ monthly, and coordinating with technicians to complete vehicle repairs on time.
- Produced detailed estimates for labor and parts, and coordinated with technicians to confirm diagnostics, repair progress, and completion timelines.
- Trained 3 new Service Advisors, providing technical guidance on shop management systems and customer handling.

SCHOLARSHIPS & AWARDS

ATCO Centennial Award	– ATCO	2025 – 2026
Jason Lang Scholarship	– University of Alberta	2024 – 2025
Speaker of the Week 5x	– UAPSSA	2024 – 2025

LEADERSHIP & COMMUNITY INVOLVEMENT

Member and Board (Nominee)

Jan 2023 – Sep 2024

CIIRSA, Fort McMurray, AB

- Delivered workshops and seminars for newcomers to Fort McMurray, covering local services, resources, and community orientation.
- Contacted CIIRSA members through outreach calls to renew annual memberships, supporting the organization's operating funding base.

ADDITIONAL INFORMATION

Willing to relocate

Interests – Gym Strength training, FPV drones, Electronics design, Hip-Hop music, Soccer playing, and Cycling.